Microprocessor to monitor weather of a town/city

Temperature
Moisture
Pressure
Light
Rainfall

INPUT:

Change in the values of Temperature, Moisture and pressure is sensed by the sensors

PROCESSING:

After the values are changed to digital using ADC,

Microprocessor will compare the values sensed with the preset/normal values.

IF value of temperature, pressure and moisture is more than normal then the value will be recorded

OUTPUT:

Graphs and charts are produced using the recorded values

Microprocessor controlled traffic signals

INPUT: Change in the value of pressure sensed by sensors

PROCESSING: After the values are changed to digital using ADC, Microprocessor will compare the values sensed with the preset or normal values stored.

If the pressure is more than normal then the signal will turn GREEN

If the pressure is less than normal then the signal turns RED

OUPUT: Signal turns REG/GREEN to control traffic

Why Microprocessor/computer is used to monitor/control conditions/situations instead of humans/manual system?

- Faster readings
- More accurate reading
- Can take frequent readings
- Can measure several physical variables at once
- No need to pay wages/salaries
- Never forget to take readings
- Never get tired
- Can work 24/7 without breaks / can work weekends
- Computers can also work in conditions where human feel unsafe